Week 43, Tuesday Session

Flipped classroom: the measures of a classifier, measure by measure – then one case: your own confusion matrix, ROC curve and AUC

Bring your laptop; open `week43tuesday.ipynb` and `exercisesweek43.ipynb`

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How today works

The format (as every Tuesday)

Mondays are lectures; Tuesdays belong to your questions and to concrete cases. Today has two parts, and the second is the skeleton of this week's exercises and of part e) of Project 2.

1. **Part A, first ∼25 min, in plenum**: Monday's worked screening example measure by measure – the four counts, accuracy and why it prefers the useless test, precision and recall and their two denominators, F_1, balanced accuracy, MCC, the AUC as a probability about pairs, and why we train on the cross entropy. Bring your questions on Monday's lecture here.
2. **Part B, the rest of the double session, in groups**: Case 1 with `week43tuesday.ipynb`: your own confusion matrix, ROC curve and AUC, written from the definitions and checked against `scikit-learn`, for logistic regression and a one-hidden-layer network from your own code on "is it an eight?" – then the `scikit-learn` calls you will use in Project 2, and the insight questions in plenum.
3. **Last 5 min**: wrap-up, the exercises, and next week.

Exercises week 43

Exercise 1: the functions of Part B, with a small test. Exercise 2: MNIST, one digit against the rest, with your Project 2 classifiers. Exercise 3 (optional): the 10×10 matrix. Deadline **Friday October 23 at midnight**; declare your use of LLMs. No validation sets yet – that is week 44.

Part A: the worked example – four counts and seven numbers

A screening test, prevalence $\pi = 2\%$, 1000 people

	predicted 1	predicted 0
true 1	$\mathrm{TP} = 15$	$\mathrm{FN} = 5$
true 0	$\mathrm{FP} = 40$	$\mathrm{TN} = 940$

Twenty people have the condition; the test finds fifteen and raises forty false alarms. Rows are the truth, columns the decision; FN are the *missed* cases (the dangerous error in screening), FP the *false alarms* (a follow-up, not a life). Next to it: the "test" that declares everybody healthy, 0, 20, 0, 980.

Ask the room first

Before showing the table: which of the two tests has the higher accuracy? Which would you rather be screened with? Why do the two answers differ?

measure	definition	test	empty
accuracy	$(\mathrm{TP}+\mathrm{TN})/n$	0.955	0.980
precision	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$	0.273	0 (0/0)
recall, TPR	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$	0.750	0
specificity, TNR	$\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$	0.959	1.000
F_1	$2\,\mathrm{prec}\cdot\mathrm{rec}/(\mathrm{prec}+\mathrm{rec})$	0.400	0
balanced acc.	$\frac{1}{2}(\mathrm{TPR}+\mathrm{TNR})$	0.855	0.500
MCC	$(\mathrm{TP}\,\mathrm{TN}-\mathrm{FP}\,\mathrm{FN})/\sqrt{D}$	0.436	0

$D = (\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})$. What each row asks: decisions right, all alike; of those flagged, how many were?; of the sick, how many found?; of the healthy, how many left alone?; both of the previous two; accuracy at equal class weights; correlation of y and $\tilde{y}$. Accuracy ranks the two tests the wrong way round; every other row ranks them the right way, and each says something different.

Part A: what the numbers mean, one at a time

Accuracy is a prevalence-weighted average

With $n_+ = \mathrm{TP} + \mathrm{FN}$, $n_- = \mathrm{FP} + \mathrm{TN}$, $\pi = n_+/n$:

$$\text{accuracy} = \frac{n_+}{n}\frac{\mathrm{TP}}{n_+} + \frac{n_-}{n}\frac{\mathrm{TN}}{n_-} = \pi\,\text{recall} + (1-\pi)\,\text{specificity}$$

$= 0.02 \times 0.75 + 0.98 \times 0.959 = 0.955$: the sick enter with weight 0.02. Compare every accuracy with the majority rule $\max(\pi, 1-\pi)$ and quote the fraction of the *remaining* error removed, $1 - (1 - \mathrm{acc})/(1 - p_0)$ – here negative. Balanced accuracy is the same average at equal weights: 0.855 against 0.5 for a coin.

Recall is about the misses, precision about the false alarms

Different denominators: the truth ($n_+ = 20$) for recall, our decision ($\mathrm{TP} + \mathrm{FP} = 55$) for precision. Three quarters of the sick are found; three of four people called back are healthy, because the condition is rare (40 false alarms among 980 healthy is a 4% false positive rate). Lower the threshold: recall up, precision down. Screening wants recall, a spam filter precision.

F_1 is small whenever either is small

With $m = \min(\mathrm{prec}, \mathrm{rec})$, $M = \max$:
$F_1 = 2mM/(m+M) = 2m/(1+m/M)$, so $m \le F_1 \le 2m$. That is why the harmonic mean: precision 1 and recall 0.1 give an arithmetic mean of 0.55 and a harmonic mean of 0.18 – perfect on one axis and useless on the other is useless. Worked example: $2 \times 0.273 \times 0.75/1.023 = 0.40$. F_1 never looks at TN.

MCC is a correlation

The Pearson correlation (Section 2.4) between the 0/1 labels and the 0/1 decisions: all four counts, symmetric in the classes, 0 for the empty rule, ± 1 perfect/perfectly wrong – hard to fool by imbalance:
$13\,900/31\,917 = 0.436$.

Why we train on the cross entropy

All of the above are functions of the decisions: staircases in the parameters, zero gradient. The cross entropy is the smooth surrogate, minimised *roughly* where the decisions are right: one positive, one negative, model A (0.51, 0.49) has accuracy 1 and cross entropy 0.673, model B (0.99, 0.60) accuracy 0.5 and 0.463.

Part A: the AUC is a probability about pairs

From one threshold to all of them

A confusion matrix is one threshold: one point $(\mathrm{FPR}, \mathrm{TPR}) = (0.041, 0.75)$ for our test. Scores give a whole curve, from $(0, 0)$ (flag nobody) to $(1, 1)$ (flag everybody); a perfect ranking passes through $(0, 1)$, a random one is the diagonal. The curve is every trade-off the model offers; a threshold picks one point.

The rank formula

Sort all $n = n_+ + n_-$ scores increasingly; let R_i be the rank of the ith positive. Below it sit $R_i - 1$ samples, $i - 1$ of them positives, so it outranks $R_i - i$ negatives. Summed over the positives: $\sum_i R_i - \frac{1}{2} n_+(n_+ + 1)$ correctly ordered pairs of $n_+ n_-$,

$$\mathrm{AUC} = \frac{\sum_{i \in \mathrm{pos}} R_i - \frac{1}{2} n_+(n_+ + 1)}{n_+ n_-} = \mathrm{Prob}(\mathrm{pos} > \mathrm{neg}),$$

the Mann–Whitney statistic (ties: average rank, half a pair), equal to the trapezoidal area under the curve.

Three checks of the formula

A random ranking: every pair is ordered correctly with probability $\frac{1}{2}$, so 0.5. All scores equal: every $R_i = (n + 1)/2$ with average ranks, $\sum_i R_i = n_+(n + 1)/2$, numerator $n_+ n_-/2$: again 0.5. Perfect separation: the positives hold the top n_+ ranks, $\sum_i R_i = n_+ n_- + \frac{1}{2} n_+(n_+ + 1)$: 1.

What the AUC ignores

The threshold – there is none in it. The prevalence – duplicate every positive: TP and FN double together, every TPR is unchanged, FPR never saw the positives. Precision is different: FP sits in its denominator, so the precision–recall curve moves up when positives are duplicated, and its floor is π – the curve to look at when positives are rare and the ROC curve flatters (1% FPR on 9108 negatives is 91 false alarms against 892 fives).

In one sentence each

Matrix: what a user of the classifier experiences at a stated threshold. AUC: what the model is capable of. A report gives both.

Case 1: your own confusion matrix, ROC curve and AUC, in five steps

Goal: leave with your own `confusion_matrix`, `classification_report`, `roc_curve` and `roc_auc`, agreeing with `scikit-learn` to every digit, and with the `scikit-learn` calls for Project 2 at your fingertips (notebook, Case 1; seed 2026). Exercises 1 and 2 are the same steps.

1. **The data and the number to beat:** the 8×8 digits, $y = 1$ for the eights (9.7%); stratified 80/20 split, 1437 training and 360 test images. "Never an eight" scores 0.9028 – write it down before training anything.
2. **Two classifiers with your own code:** logistic regression (the week-41 network with no hidden layer, $64 \to 1$, 65 parameters) and one hidden layer of 32 ReLU units (2113 parameters), SGD with $M = 32$, $\gamma = 0.1$, 150 epochs.
3. **The confusion matrix by hand:** four counts for $p > 0.5$, then accuracy, precision, recall, specificity, F_1, balanced accuracy and MCC from the definitions of Part A – checked against `sklearn.metrics` (mind its layout).
4. **The ROC curve and the AUC from one sorted pass:** the curve from cumulative TP and FP down the sorted scores, the area by the trapezoidal rule and by the rank formula; the histogram of the output probabilities by class; all against `roc_auc_score` and `roc_curve`.
5. **The same with** `scikit-learn`: `confusion_matrix`, `classification_report`, `roc_curve`, `roc_auc_score`, `ConfusionMatrixDisplay`, `RocCurveDisplay` – the template for Project 2.

Case 1: the code you should end up with

```
def confusion_matrix(y, yhat):
    tp = np.sum((y == 1) & (yhat == 1))
    fn = np.sum((y == 1) & (yhat == 0))
    fp = np.sum((y == 0) & (yhat == 1))
    tn = np.sum((y == 0) & (yhat == 0))
    return tp, fn, fp, tn

def roc_curve(y, s):
    order = np.argsort(-s, kind="mergesort")
    y, s = y[order], s[order]
    tp = np.cumsum(y == 1)
    fp = np.cumsum(y == 0)
    last = np.r_[np.diff(s) != 0, True]
    tpr = np.r_[0.0, tp[last] / tp[-1]]
    fpr = np.r_[0.0, fp[last] / fp[-1]]
    return fpr, tpr, np.r_[np.inf, s[last]]

def roc_auc(y, s): # P(pos > neg)
    _, inv, cnt = np.unique(
        s, return_inverse=True, return_counts=True)
    first = np.cumsum(cnt) - cnt + 1
    ranks = (first + (cnt - 1) / 2)[inv]
    n_pos = np.sum(y == 1)
    n_neg = len(y) - n_pos
    S = np.sum(ranks[y == 1])
    S -= n_pos * (n_pos + 1) / 2
    return S / (n_pos * n_neg)
```

```
# step 5: the scikit-learn template for Project 2
from sklearn.metrics import (
    confusion_matrix, classification_report,
    precision_score, recall_score, f1_score,
    roc_curve, roc_auc_score,
    ConfusionMatrixDisplay, RocCurveDisplay)

y_pred = p_test > 0.5
cm = confusion_matrix(y_test, y_pred)
print(cm) # [[TN, FP], [FN, TP]]
print(classification_report(y_test, y_pred))
fpr, tpr, thr = roc_curve(
    y_test, p_test, drop_intermediate=False)
print(roc_auc_score(y_test, p_test))

fig, ax = plt.subplots(1, 2, figsize=(9, 3.6))
ConfusionMatrixDisplay(cm).plot(ax=ax[0])
RocCurveDisplay.from_predictions(
    y_test, p_test, ax=ax[1])
```

Checkpoints: your ROC curve has 361 points and equals `roc_curve(..., drop_intermediate=False)`; the three AUCs agree to five decimals.

Case 1: what you should find

Steps 2–4: the test set at $t = 0.5$

	TP	FN	FP	acc	prec	rec	AUC
never an eight	0	35	0	0.903	–	0	0.5
logistic regr.	25	10	3	0.964	0.893	0.714	0.9613
one hidden layer	29	6	2	0.978	0.935	0.829	0.9896

TN = 322 and 323 of 325. The accuracies are 0.96 and 0.98 against the free 0.90: the two classifiers removed 63% and 77% of the remaining error. Logistic regression misses 10 of 35 eights; F_1 0.79 against 0.88, balanced accuracy 0.85 against 0.91, MCC 0.78 against 0.87. The hand-made numbers agree with `scikit-learn` to every digit printed; the hand-made ROC curve has 361 points and is identical to `roc_curve(..., drop_intermediate=False)`; the three AUCs agree to five decimals.

Reading it with Part A

Accuracy $= 0.097 \times \mathrm{recall} + 0.903 \times \mathrm{specificity}$: for logistic regression $0.097 \times 0.714 + 0.903 \times 0.991 = 0.964$ – the missed eights cost it 0.03, the three false alarms 0.008. The two classifiers differ almost only in FN (10 against 6): recall is the measure that moves ($0.71 \to 0.83$), precision barely ($0.89 \to 0.94$).

What the AUC adds

Logistic regression at 0.5 misses 10 eights, yet its AUC is 0.96: their p runs from 0.006 to 0.45, and each of them is still ranked above 157 to 321 of the 325 non-eights. A lower threshold would find some of them at the price of false alarms – the operating point can be slid along the curve; the histogram of step 4 shows what it would cost. Which point to choose, and on which data, is next week.

Step 5

`classification_report` prints precision, recall and F_1 for *both* classes plus their macro and weighted averages: 0.94/0.83/0.88 for the eights, 0.98/0.99/0.99 for the rest – the same numbers, with the negatives' side made explicit.

Case 1: do you have the insight?

1. **Accuracy.** Both classifiers scored above 0.95 while "never an eight" scores 0.90. What fraction of the *remaining* error did each remove? Write each accuracy as $\pi\,\mathrm{recall} + (1-\pi)\,\mathrm{specificity}$ and say which term carries it. Which single numbers survive the imbalance?
2. **The two errors.** The two classifiers differ mostly in one of the four counts. Which one, what does it mean for an eight, and which measure is it that moves most between the two?
3. **The AUC.** State in one sentence, about pairs of images, what an AUC of 0.99 means. Why does the ROC curve ignore the threshold and the class balance, and why does the precision–recall curve not?
4. **What a confusion matrix cannot say.** The AUC of logistic regression is 0.96, not 0.5, although it misses ten eights at $t = 0.5$. What does that tell you about the choice of threshold, and why is the matrix at 0.5 nevertheless what Project 2 asks for?

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What we hope you say

(1) With p_0 the majority accuracy, the fraction removed is $1-(1-\mathrm{acc})/(1-p_0)$: 63% and 77%. The accuracy is $0.1\times\mathrm{recall} + 0.9\times\mathrm{specificity}$, carried almost entirely by the specificity, so missing a quarter of the eights still scores above 0.95; balanced accuracy, F_1 and the MCC give the majority rule 0.5 or 0 and do not reward it. (2) FN, the missed eights; recall is the measure that moves, $0.71 \to 0.83$. (3) Pick a random eight and a random non-eight: with probability 0.99 the eight gets the higher score. The ROC curve is built from the ranking alone, and copying every eight leaves every rate unchanged; precision has FP in its denominator, so the fraction of positives enters it. (4) 0.5 is a convention; a lower threshold would find more of the eights at the price of false alarms – the operating point can be moved along the curve, and choosing it, on validation data and from the costs of the two errors, is next week. The matrix at a stated threshold is what a user of the classifier experiences; the AUC is what the model is capable of. A report gives both.

Wrap-up, the exercises, and next week

What you should own now

A confusion matrix you can compute and read without a library; accuracy, precision, recall and F_1 from their definitions and from the table of Part A, with the two identities ($\mathrm{acc} = \pi\,\mathrm{rec} + (1-\pi)\,\mathrm{spec}$, $m \le F_1 \le 2m$); the ROC curve and the AUC from a sorted pass and from the rank formula, checked against `scikit-learn`; and the `scikit-learn` calls for Project 2.

Exercises week 43 – deadline Friday October 23 at midnight

`exercisesweek43.ipynb`: exercise 1 is steps 3–5 as functions with a small test (tied scores on purpose); exercise 2 is steps 1–4 on MNIST, one digit against the rest, with your Project 2 classifiers; exercise 3 (optional) is the 10×10 matrix. Hand in one notebook per group in Canvas; list your collaborators; finish the self-check table; declare your use of LLMs.

Project 2 (deadline November 6) and next week

Part e) of Project 2 asks for logistic regression as the baseline and for the confusion matrix on the full MNIST test set with its interpretation: today's steps 3–5 on the full data, read with Part A. **Week 44, Tuesday session and exercises:** validation sets and cross-validation for your Project 2 network – choosing the hyperparameters, and the threshold, without touching the test set. **Week 44, Monday:** Chapter 10 continued – equivariance and pooling.